

Dynamics - Technical Project

Project 2. – “Projectile”

Objective:

A series of technical projects provide students with a hands-on experience to understand fundamental concepts in Dynamics through “learning by doing.” Throughout the ME 2330 course lectures, students will learn about basic principles, concepts, analytical approaches, and problem-solving skills in Dynamics. To supplement theories and “learning by lectures and problem-solving,” this project will help students connect the concepts and theories in class with experiments and “doing” as a group.

Summary:

This project is to run a projectile by a simple experiment. A group of 3~4 students comes up with their strategy to find an initial velocity.

$$x = v \cdot \cos(\theta) \cdot t \quad (1)$$

$$y = v \cdot \sin(\theta) \cdot t - \frac{1}{2}gt^2 \quad (2)$$

Your projectile could be either symmetric or asymmetric. You launch or throw an object with a known initial angle, θ , and then you measure the travel time and x-directional distance. In Equation (1), the only unknown variable is the initial velocity, v .

Here, we assume that there is no wind effect, drag, or other existing external factors, as well as the object’s mass.

Submission Guideline:

1. Write (type) a one or two-page document.
2. Record a short (less than 30 seconds) YouTube video.
3. The video shows the motion of your projectile.
4. No cover page is necessary.
5. Take a group photo.
6. Submit 1) a document, 2) a video link, and 3) a group photo through iLearn.

Measurement Guideline:

1. Your group can pick any object.
2. At least 2 or 3 experiments are recommended for accuracy.
3. Work as a group. (Each individual plays a specific role)
4. Always be careful! (Safety is the first)
5. Brainstorm and discuss first before you go out for an experiment.